

JAPKEERAT KAUR

AI Engineer, Data Science Researcher

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PROFESSIONAL SUMMARY

Dynamic AI Engineer and Data Science Enthusiast with hands-on expertise in deploying LLMs, NLP, and complex deep learning architectures. Combines a strong technical foundation in programming and vector databases with proven executive leadership managing cross-functional teams and large-scale event logistics. Adept at applying data-driven strategies and intelligent automation to solve complex technical and operational challenges.

WORK EXPERIENCES

AI Intern | Internship Jan 2024 - June 2024
Tech Mahindra Ltd. On-site • Hinjewadi, Pune

- Excelled in Large Language Models (LLMs) and Natural Language Processing (NLP) along with vector databases.
- Improved an advanced Retrieval-Augmented Generation (RAG) pipeline utilizing Llama Index, enhanced by a Reinforcement Learning from Human Feedback (RLHF) training pipeline to improve model accuracy and contextual relevance.

Technologies / Skills Used : LLMs, NLP, Cloud Storage Databases, Llama Index, RLHF, RAG

Web Developer May 2024 - Jun 2024
RestoryaPhysio Pune

- Designed and created a fully responsive, custom website for a physiotherapy clinic utilizing HTML and React to establish a robust professional digital presence.
- Delivered an intuitive and accessible user interface tailored strictly to the client's healthcare requirements, streamlining patient access to services and enhancing online engagement.

Technologies / Skills Used : HTML, React

PROJECTS

Digital Subscription and Tool Optimizer | GitHub Jun 2026 - Present
Solo Project

- Designed a local RAG application that scrapes the GitHub API to dynamically map expensive software subscriptions to free open-source alternatives.
- Built a semantic search engine using Weaviate and BGE-M3 vector embeddings to accurately match user budget constraints with relevant repository capabilities.
- Deployed Llama 3 via Dockerized Ollama to synthesize retrieved data into automated, offline cost-mitigation reports with zero external API runtime costs.

Technologies / Tools Used : Python, RAG, LLM, Weaviate, Docker, Ollama, GitHub API, HuggingFace

IOT and Adaptive RL Based Effluent Water Treatment Plant | Working of Prototype Sep 2025 - Present
Backend Developer

- Developed an IoT-enabled Industrial Wastewater Treatment prototype, integrating real-time sensor arrays (pH, temperature, turbidity) to continuously monitor water quality.
- Implemented Reinforcement Learning (RL) algorithms to analyze sensor data streams and autonomously optimize filtration and treatment processes.
- Created an end-to-end smart automation system, applying intelligent control loops to enhance industrial resource efficiency and sustainability.

Technologies / Tools Used : RL, Arduino UNO, Blynk

Identification of Medicinal Plants and Informing About Which Diseases Can Be Cured By Them Sep 2023 - Oct 2024
Backend Developer

- Engineered and benchmarked various Deep Learning architectures, including CNN, LSTM, and Bidirectional LSTM (Bi-LSTM), individually and as hybrid networks to maximize prediction accuracy.
- Integrated hybrid neural networks that combined the spatial feature extraction of CNNs with the sequential, long-term dependency tracking of Bi-LSTMs for improved data modeling.
- Conducted rigorous iterative testing and performance tuning, analyzing model evaluation metrics to optimize hyperparameters, reduce overfitting, and identify the most robust deployment candidate.

Technologies / Tools Used : Python, HTML, Gemini API, CNN, LSTM, ML

SKILLS

AI & Machine Learning :	RLHF, LlamaIndex, HuggingFace, LLMs, NLP, DL, RL, RAG
Programming Languages :	C, C++, Java, Python, JavaScript, SQL, HTML
Databases :	PostgreSQL, MongoDB, MySQL, WeAviate
Frameworks & Libraries :	React, Node.js, FastAPI, Spring Boot, Next.js
Tools & Platforms :	Excel, Power BI, Tableau, GitHub, Git, Docker, Postman
Soft Skills :	Leadership, Time Management, Analytical Skills, Communication, Problem Solving, Critical Thinking
Languages :	English, Hindi

EDUCATIONS

MIT World Peace University | Kothrud, Pune Jul 2024
B. Tech Computer Science and Engineering (Artificial Intelligence and Data Science) CGPA : 9.04

MIT World Peace University | Kothrud, Pune Aug 2021 - Jun 2024
Diploma in Computer Science and Engineering (Artificial Intelligence and Data Science) CGPA : 9.96

EXTRACURRICULAR ACTIVITIES

Wheaton Warrenville South High School, Wheaton, Illinois, USA 2019 - 2020

- Part of Wheaton Warrenville South High School Varsity Badminton Team
- Part of Math Team, Won 1st Place for district tournament (Group) and 2nd Place for individual tournament
- Contributed to Windows systems in Cyber Patriots Club.
- Awarded a coin for achievements in Cyber Patriots.

AWARDS & ACHIEVEMENTS

Internal Smart India Hackathon Finalist (2023, 2024, 2025)
MIT World Peace University, Kothrud, Pune
[Certificates Link](#)

HackMITWPU 2024 Workathon Winner
MIT World Peace University, Kothrud, Pune
[Certificate Link](#)